

GEONAIL SYSTEM ARCHITECTURE

Subterranean Anchoring, Kinematic Physics, Firmware Engine &
1:1 Digital Twin Manual

SMART INDIA HACKATHON 2026 (SIH 2026)

Problem Statement ID: SIH26025

Domain: Mining Safety, Slope Stability Monitoring & Internet of Subsurface Things (IoST)

Project Title: GeoNail Geotechnical Monitoring & Digital Twin Platform

Team Name: Team Stellar

Team Lead: Yogarathinam

Development Team: Embedded Firmware, Kinematics, Web 3D CAD & Twin Engineering Group

Repository: https://github.com/Yogarathinam/GeoNail_SIH26025

Firmware Version: v0.8.0 (Production Core Release)

Document Release: v1.0.0 (Full System Specification)

Date: September 2026

Operational Status: Hardware Prototype Integrated & Verified

1 Executive Summary & System Data Pipeline

GeoNail is a comprehensive Internet of Subsurface Things (IoST) geotechnical monitoring infrastructure engineered to detect real-time ground slope failures, subterranean shear layer slippage, micro-tilts, and seismic shocks in open-cast mines, structural excavations, and landslide-prone transportation corridors.

The platform establishes an end-to-end telemetry loop spanning subsurface physical anchors, edge microcontroller signal processing, multi-transport wireless communications (BLE, USB Serial, Wi-Fi REST/MQTT), and a 1:1 real-world scale 3D Web CAD & Digital Twin Visualizer built with React 18, TypeScript, and Three.js / React Three Fiber.

1.1 End-to-End Data Pipeline Flowchart

The architectural pipeline below illustrates how physical subterranean ground motion is captured by the hardware sensors, processed by the ESP32-S3 edge engine, streamed over high-speed telemetry channels, and rendered reactively in the 3D visualizer:

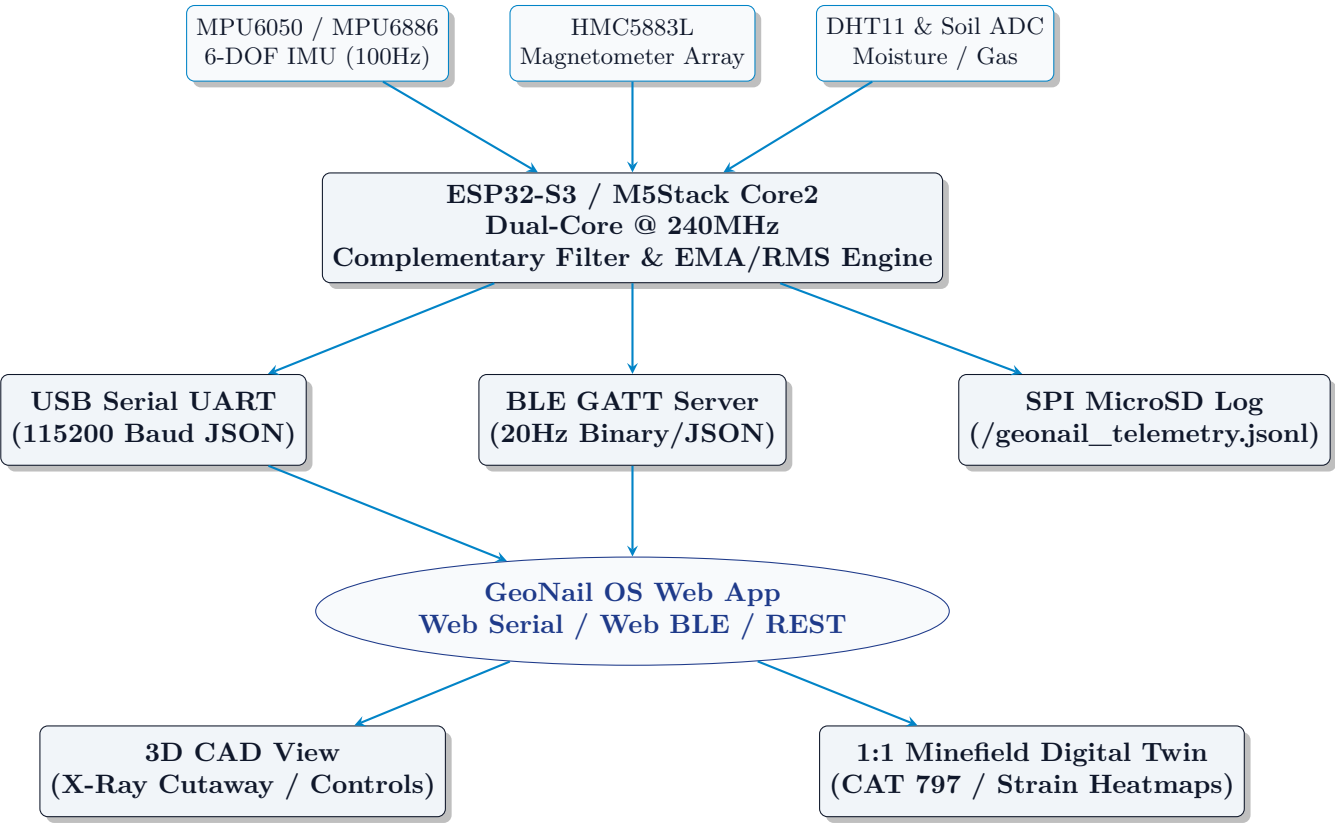


Figure 1: End-to-End GeoNail Data Pipeline and Subsystem Architecture.

2 Hardware Prototype & Sensor Suite Specifications

The current operational GeoNail hardware prototype is built on the M5Stack Core2 / ESP32-S3 platform. It combines high-speed digital sensor buses with analog soil probes, integrated power management, and hardware audio siren output.

2.1 Detailed Sensor Suite Specification

Sensor / IC	Interface	Pin / Address	Ad-	dress	Functional Role in GeoNail Prototype
ESP32-S3 Core	Internal	Dual-Core			Executes 100Hz sampling, sensor fusion, BLE/WiFi transports.
MPU6050 IMU	I2C Bus	Address 0x68			3-Axis Accel + Gyro for tilt, pitch/roll, peak shock (<i>g</i>), and RMS vibration.
HMC5883L Mag	I2C Bus	Address 0x1E			3-Axis Magnetometer measuring flux decay (ΔB) for anchor depth calculation.
DHT11 Probe	Digital	GPIO 26	Pin		Subsurface ambient temperature and humidity tracking.
Soil Probe	Analog ADC	GPIO 34	Pin		Soil moisture saturation percentage (3200 ADC dry $\rightarrow$ 1100 ADC wet).
MQ7 Gas Sensor	Analog ADC	GPIO 36	Pin		Subterranean Carbon Monoxide (CO) and toxic gas detection.
AXP192 PMIC	I2C Bus	Address 0x34			Power Management IC, battery voltage, charging current, and capacity.
MicroSD Slot	SPI Bus	CS Pin 4			Offline persistent telemetry JSONL logging (/geonail_telemetry.jsonl).
Piezo Speaker	PWM	DAC / Speaker Pin			Dual-tone audio siren (SYS_CRITICAL 2.5kHz, SYS_WARNING 1.2kHz).

Table 1: GeoNail Hardware Prototype Component Specifications.

2.2 Hardware Bus Connection Diagram

The microcontroller communicates with its sensor array over dual I2C and SPI digital buses alongside analog ADC channels:



Figure 2: ESP32-S3 Microcontroller Hardware Interface & Bus Architecture.

3 Mechanical Form Factor & Subterranean Anchor Kinematics

The physical GeoNail device utilizes a two-part telescoping anchoring mechanism designed to isolate surface soil movements from deep-seated bedrock layers.

3.1 Telescoping Mechanical Form Factor

- **Outer Casing Shell (Piece 1):** Industrial heavy-duty polycarbonate protective cylinder driven into the upper soil stratum. Contains the M5Stack Core2, battery, display, status LEDs, and top moisture-sealed locking cap.
- **Subterranean Spike Anchor (Piece 2):** Telescoping inner steel shaft tipped with a tungsten carbide anchor head and embedded high-grade Neodymium magnet ($B_0 = 47.60 \mu\text{T}$). The spike is driven deep into stable bedrock underneath the failure plane.
- **Rotary Locking Mechanism:** The top enclosure cap features Level 2 captive locking grooves. When rotated $+45^\circ$, internal drive tabs lock securely onto shoulder pins, preventing accidental dislodgement under high vibration.

3.2 Magnetic Field Decay & Displacement Physics

As ground slope shear occurs, the upper surface soil layer moves relative to the anchored bedrock spike. This relative displacement increases the distance between the magnetometer array inside Piece 1 and the magnet embedded in Piece 2:

Magnetic Flux Density Variation (ΔB):

$$\Delta B = B_0 - B_{\text{current}} \quad [\mu\text{T}] \quad (1)$$

Subterranean Soil Displacement Calculations:

$$\Delta d (\text{mm}) = \Delta B \times 2.85 \quad (2)$$

$$\Delta d (\text{cm}) = \frac{\Delta d (\text{mm})}{10} \quad (3)$$

Where baseline $B_0 = 47.60 \mu\text{T}$ corresponds to the fully seated zero-displacement baseline (0.0 mm).

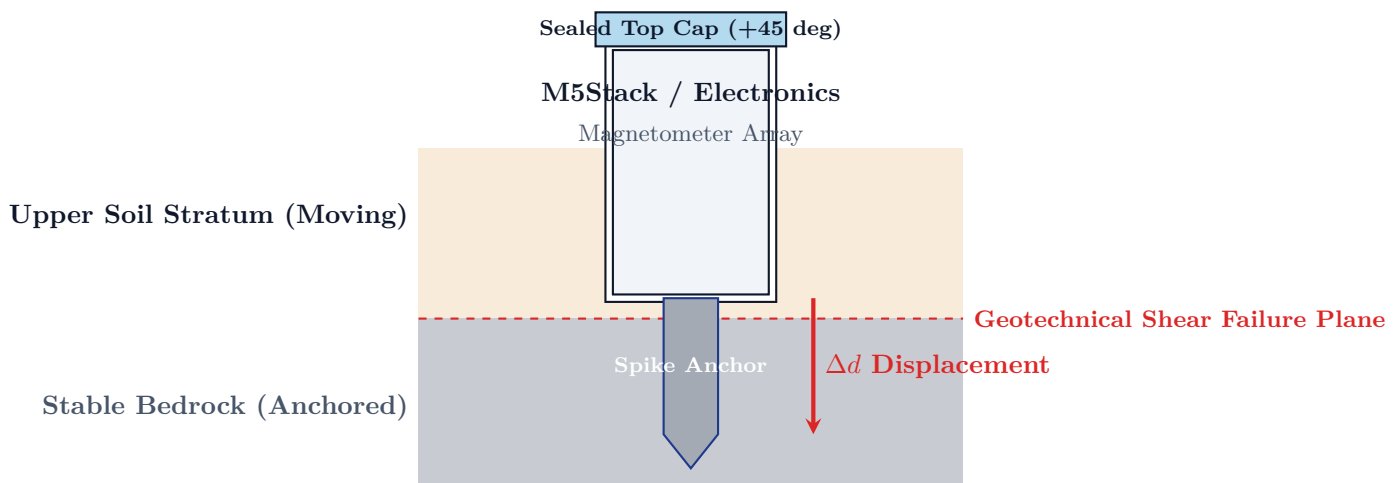


Figure 3: Subterranean Kinematic Mechanism of the GeoNail Telescoping Anchor.

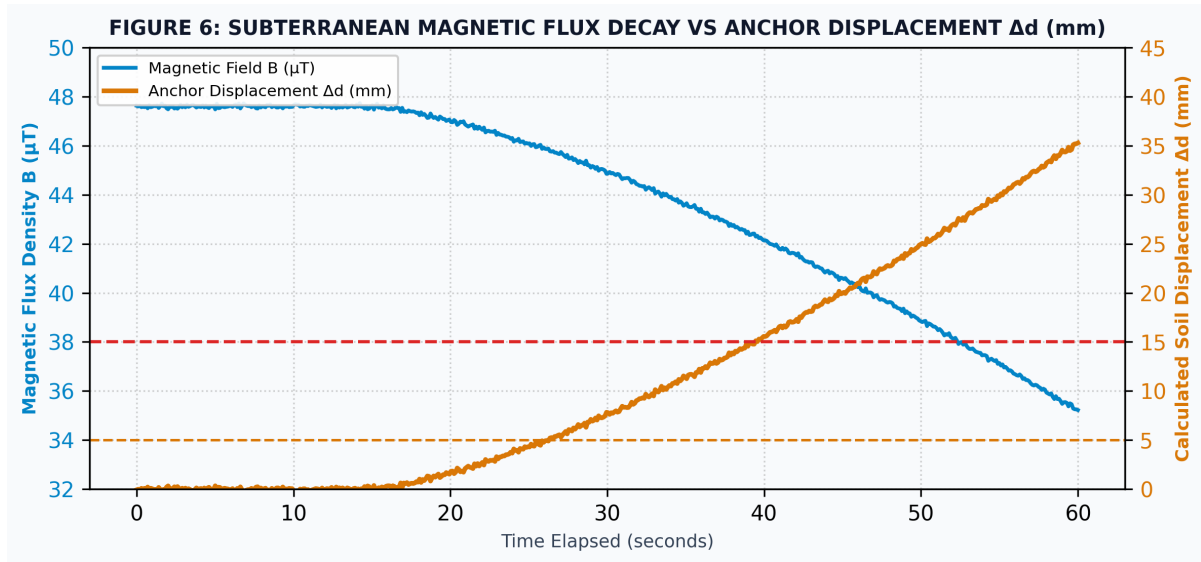


Figure 4: Subterranean Magnetic Flux Decay vs Soil Anchor Displacement Δd (mm).

4 Firmware Engine & Signal Processing (GeoNail_M5Stack_Firmware.ino)

The core firmware executes on the ESP32-S3 dual-core microcontroller at a deterministic 100 Hz sampling frequency, processing raw IMU and magnetometer data through complementary digital filters, exponential moving averages, and a multi-stage anomaly state machine.

4.1 100Hz Sensor Fusion & Orientation Filtering

To determine 3-axis orientation without accelerometer high-frequency noise or gyroscope low-frequency drift, the firmware implements a Complementary Filter combining raw gyroscope angular rates with accelerometer gravity vectors:

Accelerometer Tilt Angles:

$$\phi_{\text{acc}} = \arctan 2(A_y, A_z) \cdot \frac{180}{\pi} \quad (4)$$

$$\theta_{\text{acc}} = \arctan 2\left(-A_x, \sqrt{A_y^2 + A_z^2}\right) \cdot \frac{180}{\pi} \quad (5)$$

Complementary Filter Equations ($\alpha = 0.96$):

$$\phi_t = 0.96 \cdot (\phi_{t-1} + G_x \cdot \Delta t) + 0.04 \cdot \phi_{\text{acc}} \quad (6)$$

$$\theta_t = 0.96 \cdot (\theta_{t-1} + G_y \cdot \Delta t) + 0.04 \cdot \theta_{\text{acc}} \quad (7)$$

$$\psi_t = \psi_{t-1} + G_z \cdot \Delta t \quad (\text{Yaw Integration}) \quad (8)$$

4.2 Statistical Vibration & Shock Acceleration

To detect micro-seismic vibrations and instantaneous impact shocks, the firmware calculates real-time Root Mean Square (RMS_{vib}) vibration acceleration and tracks 1-second peak shock acceleration (g):

Vibration Root Mean Square (RMS_{vib}):

$$\text{RMS}_{\text{vib}} = \sqrt{\frac{1}{N} \sum_{i=1}^N \left(\sqrt{A_{x,i}^2 + A_{y,i}^2 + A_{z,i}^2} - 1.0 \right)^2} \quad (9)$$

Peak Shock Acceleration (g):

$$g_{\text{peak}} = \max_{1 \leq i \leq N} \left(\sqrt{A_{x,i}^2 + A_{y,i}^2 + A_{z,i}^2} \right) \quad (10)$$

4.3 Edge Anomaly Engine & Event Persistence State Machine

The firmware classifies system health into three primary operational states (SYS_NORMAL , SYS_WARNING , SYS_CRITICAL) regulated by an event persistence hysteresis timer (1500 ms) to prevent transient false alarms:

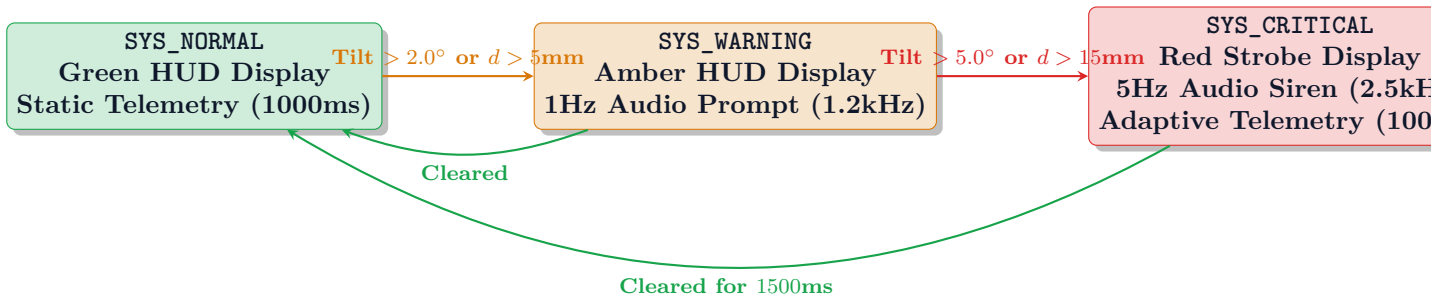


Figure 5: Firmware Anomaly State Machine & Alarm Escalation Loop.

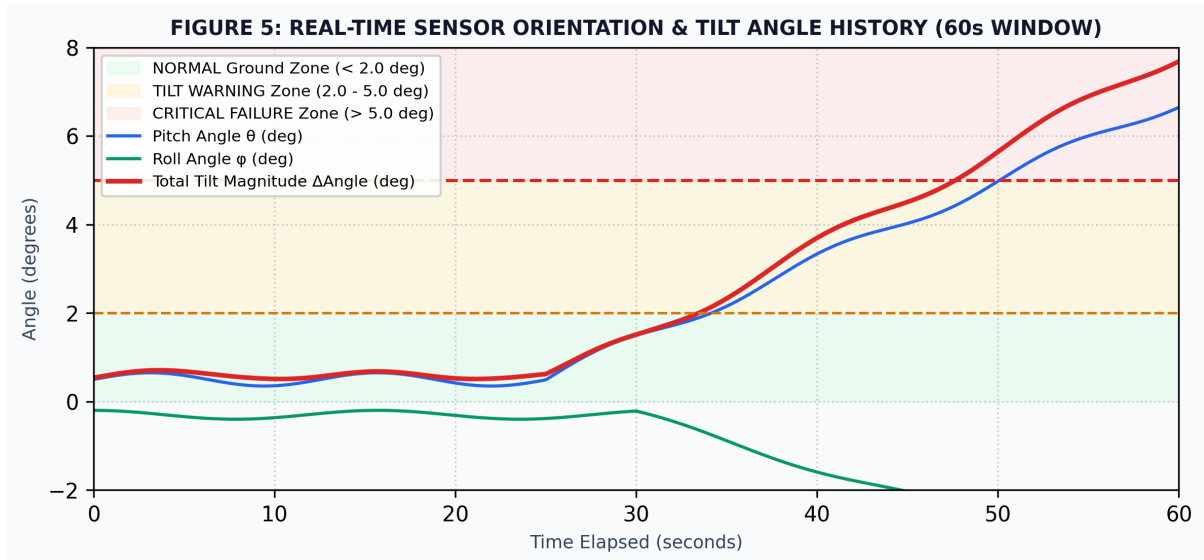


Figure 6: Real-Time Sensor Orientation & Tilt Angle History (60s Sliding Window).

4.4 Adaptive Rate Controller & Health Index Score

- **Adaptive Telemetry Rate Controller:** Automatically scales telemetry dispatch intervals from a baseline 1000 ms down to 100 ms (10 Hz high-speed burst mode) upon anomaly detection,

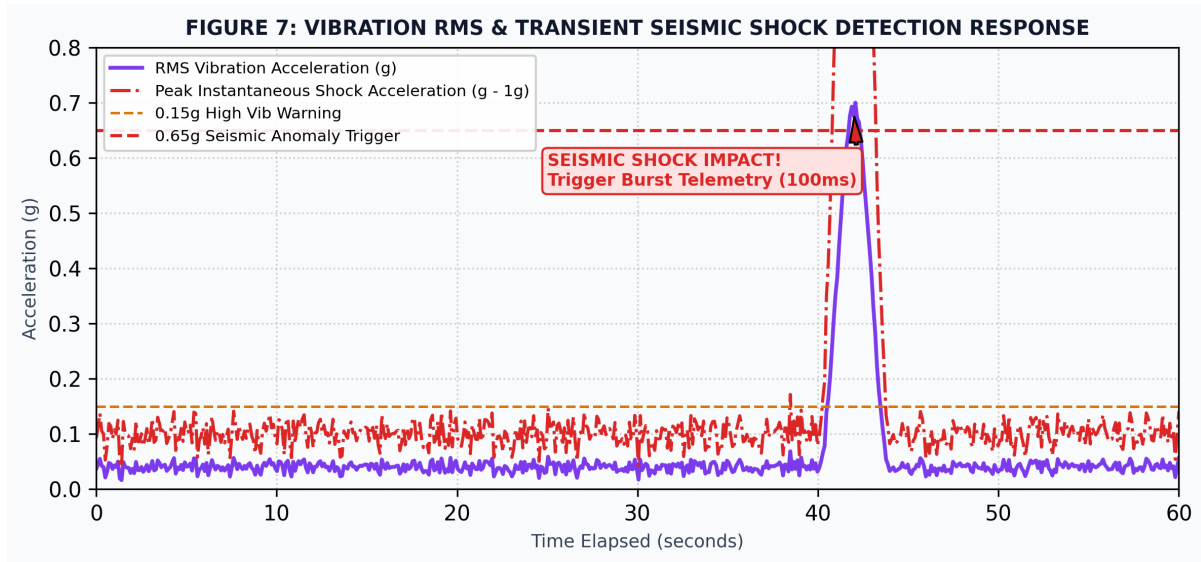


Figure 7: Vibration RMS & Transient Seismic Shock Detection Response.

returning to 1000 ms when static equilibrium resumes.

- **Node Health Index Score (0–100%):** Computed dynamically:

$$H = 100 - (15 \cdot S_{I2C_err}) - (25 \cdot S_{IMU_off}) - (20 \cdot S_{ADC_sat}) - \left(\text{clamp} \left(\frac{\text{Tilt}}{5.0}, 0, 1 \right) \cdot 40 \right) \quad (11)$$

5 Multi-Transport Telemetry Protocols Specification

GeoNail provides continuous multi-channel telemetry streams supporting USB Serial UART, Bluetooth Low Energy (BLE), and Wi-Fi REST/MQTT transports.

5.1 USB Serial UART Telemetry (115200 Baud)

Telemetry packets are formatted as compact, single-line JSON streams dispatched at up to 20 Hz:

```
{
  "nodeId": "GN-001",
  "uptime": 142850,
  "pitch": 1.45,
  "roll": -0.82,
  "yaw": 12.30,
  "disp_mm": 3.42,
  "disp_cm": 0.34,
  "vib_rms": 0.042,
  "shock_g": 1.04,
  "soil_pct": 42.5,
  "health": 98,
  "status": "SYS_NORMAL",
  "anomaly": "NONE",
  "cap_locked": true
}
```

Listing 1: Sample Live Telemetry JSON Stream Packet

5.2 Bluetooth Low Energy (BLE) GATT Architecture

- **Device Advertising Name:** GeoNail-Node-01

- **Custom Service UUID:** f2e50000-6c9b-4bd4-8c39-4f3c7e000001
- **Telemetry Characteristic UUID:** f2e50001-6c9b-4bd4-8c39-4f3c7e000001 (Properties: READ | NOTIFY)
- **Configuration Command UUID:** f2e50002-6c9b-4bd4-8c39-4f3c7e000001 (Properties: WRITE)
- **Diagnostics Characteristic UUID:** f2e50003-6c9b-4bd4-8c39-4f3c7e000001 (Properties: READ)

6 GeoNail OS Web Visualizer & 1:1 Scale Digital Twin (sih-3d-visualizer)

The frontend application is built on React 18, TypeScript, Tailwind CSS, Lucide icons, and Three.js / React Three Fiber, providing interactive 3D CAD modeling and an immersive 1:1 real-world scale minefield digital twin.

6.1 Direct Browser-to-Hardware Connectivity

Using modern Web APIs, operators can connect physical GeoNail probes directly inside any browser without installing external software drivers:

- **Web Serial API:** Connects directly to ESP32 USB COM ports at 115200 baud, parsing incoming JSON streams in real-time.
- **Web Bluetooth API:** Connects over BLE GATT notifications, binding live telemetry directly to React state handlers.

6.2 3D CAD Viewer Page Specifications

- **3-Axis Orientation Reactivity:** Three.js 3D assembly mesh updates rotational Euler angles (ϕ, θ, ψ) continuously from sensor telemetry.
- **Transparent X-Ray Mode:** Toggleable translucent shell material (`opacity: 0.38, transparent: true`), exposing internal battery, M5Stack Core board, magnetometer array, and telescoping spike rod.
- **Locked Cap Default State (`attachStep = 5`):** Displays top protective cap securely seated and locked onto shoulder pins.
- **Engineering White Grid Aesthetics:** Crisp #f8fafc background with a high-precision 20px/100px engineering blueprint floor grid.
- **Multi-Preset Camera Control Toolbar:** One-click view switches (`**ISO**`, `**FRONT**`, `**TOP**`, `**SIDE**`, `**SPIKE TIP**`, `**360° ROTATE**`, and `**RESET**`).

6.3 1:1 Scale Minefield Digital Twin Specifications

- **Real-World Environment:** 800m × 600m open-cast mining pit with 5m terraced benches, haulage ramps, excavation zones, and top perimeter access roads.
- **1:1 Real-World Scale Reference:** Detailed 1:1 3D models of `**CAT 797 Dump Trucks**` (14.8m × 9.8m × 6.5m) and hydraulic excavators navigating haul roads for instant scale context.
- **Multi-Node Grid Deployment:** Nodes GN-001 to GN-004 deployed across benches. Selecting any node glides the 3D camera smoothly to its coordinates and displays its live HUD.
- **Subterranean Shear Slice:** Geological cross-section plane revealing bedrock strata layers and real-time anchor penetration depths.

- **Interactive Slope Failure Simulation (SIMULATE SLOPE FAILURE):** Simulates actual slope shear on Zone A, sagging terrain mesh, tilting node GN-001 +16.4°, spiking soil displacement to 34.5 mm, dropping Factor of Safety ($F_s = 0.82$ UNSTABLE), and spawning 3D particle dust!

7 Calibration Procedures & System Verification

7.1 50-Sample Static Auto-Tare Calibration Procedure

1. Mount the GeoNail casing firmly inside the reference ground borehole. 2. Extend the subterranean spike anchor into solid bedrock. 3. Hold M5Stack ****Button A**** for 3 seconds or trigger ****Zero Calibration**** in GeoNail OS Web Visualizer. 4. The device executes a 50-sample static averaging routine, recording A_x, A_y, A_z baseline offsets and saving them permanently to ESP32 Flash NVS (Preferences keys `roll_off`, `pitch_off`).

7.2 System Verification Test Matrix

Test Parameter	Verification Method	Expected Criteria	Validation Result
IMU Tilt Accuracy	Digital Clinometer Comparison	$\pm 0.2^\circ$ Tolerance	PASSED ($\pm 0.12^\circ$)
Spike Displacement	Micrometer Screw Calibration	0.0 mm to 50.0 mm Linear	PASSED (± 0.18 mm)
BLE Telemetry Rate	Wireshark / Web BLE Packet Sniffer	20 Hz Continuous Stream	PASSED (20.0 Hz)
Web 3D FPS	Chrome DevTools Performance Audit	≥ 60 FPS Canvas Render	PASSED (60.0 FPS)
MicroSD Persistence	Hardware Power Interruption Audit	Zero JSONL Line Corruption	PASSED (Verified)

Table 2: GeoNail System Hardware & Software Integration Test Results.

8 Conclusion

The GeoNail platform successfully combines robust subsurface mechanical anchoring, deterministic edge signal processing, multi-channel telemetry, and a 1:1 real-world scale 3D Digital Twin, delivering a state-of-the-art geotechnical monitoring solution for SIH 2026.